

LULU SMOOTHERS ON ONLINE DATA

Willie J. Conradie

Department of Statistics and Actuarial Science, University of Stellenbosch,
Private Bag X1, 7602 Matieland, South Africa

and

Tertius De Wet

Department of Statistics and Actuarial Science, University of Stellenbosch,
Private Bag X1, 7602 Matieland, South Africa

Key words: Compound LULU smoother, Gaussian noise, Spiky noise, Outliers, Level shift, Trend

Summary: This study investigates the ability of the compound LULU smoother to reduce Gaussian and spiky impulsive noise and preserve level shifts in case of a constant signal and in a trend period in online data. This smoother belongs to the class of LULU smoothers that was introduced into the mathematical literature by Rohwer (1989). Monte Carlo experiments for a comparison of the compound LULU smoother with the running median in basic data situations and in the presence of observational noise were carried out. It was found that the compound LULU smoothers perform extremely well in the cases when the level shift or outliers are in the opposite direction of the trend.

1. Introduction

The smoothing of online monitoring data is an important task, for example in the medical field important life-related decisions are dependent on the outcome of such data. The heart rate and blood pressure of patients under observation are measured at short time intervals and these data provide important information for medical decision support. The basic goal is to extract

AMS: 37M10, 62G05

the underlying, clinically relevant signal from the observed time series. Short-term fluctuations and outliers caused by, for example, the movement of the patient or measurement errors, should be removed, while sudden level shifts and monotonic trends should be preserved. Medical researchers should know which length of blockpulse of blood pressure can be ignored, or is dangerous to the patient, and hence the smoothing algorithm can be set to correspond to these limits.

Fried, Bernhold and Gather (2006) investigated robust hybrid filters which are designed to combine the good properties of finite impulse response median hybrid (FMH) filters of Heinone and Neuvo (1987, 1988) with the robustness of the repeated median in trend periods. To determine how successful these filters will be in the smoothing of online medical data, Fried et al. (2006) investigated analytical properties of the resulting hybrid filters and compared their ability to attenuate Gaussian noise, to resist outliers and to preserve abrupt shifts in case of a constant signal and in a trend period via simulations.

In this paper the compound LULU smoother (cf Conradie, De Wet and Jankowitz, 2009) is investigated to determine its ability to smooth online data successfully under circumstances considered in the Fried et al. (2006) paper. This smoother belongs to the class of LULU smoothers that was introduced into the mathematical literature by Rohwer (1989). Rohwer (1989, 1999, 2002a, 2002b, 2005) and Rohwer and Toerien (1991) in a series of papers, showed that they have very attractive mathematical properties. These include idempotency, co-idempotency, stability, trend preserving and variation decomposing properties which make them worthy competitors to the well known nonlinear smoothers found in the literature. In Conradie De Wet and Jankowitz (2005, 2006) some practical and statistical aspects of LULU

smoothers were investigated. In Conradie et al. (2009) a number of LULU smoothers were compared with a number of Tukey's median smoothers in their ability to recover a sinusoidal signal of different frequencies with contaminated normal noise and impulsive noise added. It was found that LULU smoothers perform substantially better than Tukey's median smoothers in recovering a sinusoidal signal when blockpulses are present in the data series.

In this paper the extent to which the compound LULU smoother reduces Gaussian and spiky impulsive noise and preserves level shifts in case of a constant signal and in a trend period, is determined. This investigation is motivated by the sound mathematical theory that underlies LULU smoothers as well as their good performance in the comparisons in Conradie et al. (2009). It investigates the unexplored performance of LULU smoothers in handling spiky noise and level shifts under the circumstances mentioned above.

The following basic components model will be used for an observed time series (x_t) :

$$x_t = \mu_t + u_t + v_t, \quad t \in \mathbb{Z}. \quad (1.1)$$

Here μ_t is the signal, that is a smoothly varying level of the series, possibly with a few abrupt shifts, u_t is symmetric observational noise with mean zero and variance σ^2 , and v_t is spiky impulsive noise from an outlier generating mechanism.

In the next section LULU smoothers are defined and a number of their most important properties are discussed. In section 3 Monte Carlo experiments to compare the compound LULU smoother with the running median in basic data situations and in the presence of observational noise, were carried out. The compound LULU smoother and the running median are applied to two practical data series in section 4 and in section 5 a number of concluding remarks are made.

2. LULU Smoothers

In this section a brief overview of LULU smoothers is given. It introduces the basic definitions and most important properties to the initiated reader.

2.1 Definitions

A class of nonlinear smoothers, called LULU smoothers, was developed by using the minimum and maximum selectors. These smoothers have very attractive properties, for example idempotency, co-idempotency, edge preservation, variation reduction, etc. (see Rohwer, 2005, for a full discussion).

A series x is defined as a doubly infinite numerical sequence:

$$x = \{\dots, x_{-3}, x_{-2}, x_{-1}, x_0, x_1, x_2, x_3, \dots\}$$

with x_i the observation at time i of a time varying random variable or phenomenon X .

LULU smoothers are compositions of the following two basic rank selectors operating on x :

$$(\vee^n x)_i = \max\{x_i, \dots, x_{i+n}\}$$

$$(\wedge^n x)_i = \min\{x_{i-n}, \dots, x_i\}$$

for $n = 1, 2, \dots$ and $i = 0, \pm 1, \pm 2, \dots$. These are called the forward maximum selector and the backward minimum selector, respectively. Note that $\wedge^n$ removes upward pulses and $\vee^n$ removes downward pulses. The first step in building LULU smoothers is to combine $\wedge^n$ and $\vee^n$ to form the upper and lower half smoothers U_n and L_n respectively, i.e.

$$(U_n x)_i \equiv (\wedge^n \vee^n x)_i = \min\{\max(x_{i-n}, \dots, x_i), \dots, \max(x_i, \dots, x_{i+n})\}$$

and

$$(L_n x)_i \equiv (\vee^n \wedge^n x)_i = \max\{\min(x_{i-n}, \dots, x_i), \dots, \min(x_i, \dots, x_{i+n})\}.$$

The next step is to use compositions of the half smoothers L_n and U_n , viz. $L_n U_n$ and $U_n L_n$, called basic smoothers, to define the class of LULU full smoothers below, which smooth simultaneously from above and below.

It can easily be proven (cf. Rohwer, 2005) that

$$(U_n L_n x)_i \leq (M_n x)_i \leq (L_n U_n x)_i$$

where $(M_n x)_i$ is the running median smoother of window size $2n + 1$, defined as

$$(M_n x)_i = \text{med}(x_{i-n}, \dots, x_i, \dots, x_{i+n}).$$

Note that $U_n L_n$ and $L_n U_n$ have a window size of $4n + 1$.

The half smoothers C_n , called ceiling, and F_n , called flooring, are recursively defined as follows:

$$C_1 = L_1 U_1 \text{ and } C_n = L_n U_n C_{n-1}$$

and

$$F_1 = U_1 L_1 \text{ and } F_n = U_n L_n F_{n-1}$$

with $C_0 = F_0 = I$.

Using C_n and F_n a Winsorised smoother can be defined as

$$(B_n^* x)_i = \begin{cases} x_i & \text{if } x_i \in [(F_n x)_i, (C_n x)_i] \\ (F_n x)_i & \text{if } x_i < (F_n x)_i \\ (C_n x)_i & \text{if } x_i > (C_n x)_i. \end{cases}$$

B_n^* can be extended to a compound LULU smoother B_n , defined as

$$(B_n x)_i = \begin{cases} (B_{n-1} x)_i & \text{if } (B_{n-1} x)_i \in [(F_n x)_i, (C_n x)_i] \\ (F_n x)_i & \text{if } (B_{n-1} x)_i < (F_n x)_i \\ (C_n x)_i & \text{if } (B_{n-1} x)_i > (C_n x)_i. \end{cases}$$

These are LULU full smoothers. B_n^* retains the original sequence between C_n and F_n , whereas B_n is a smoother where that part of the sequence between C_n and F_n , which was not smoothed by B_m , $m < n$, is preserved. If $(F_n x)_i \leq (B_{n-1} x)_i \leq (C_n x)_i$ $(B_n x)_i = (B_{n-1} x)_i$ and it is considered as signal plus well-behaved noise. In previous studies (cf Conradie et al., 2009)

it was found that the compound smoother performs very well when compared with other non-linear smoothers. Intuitively it is also attractive since it peels off variation, that can be monitored, as n increases. This property will be explained below and practically illustrated in section 4. Hence, in this paper the focus is on this compound smoother.

The successive elements in a sequence that generate one output value of a smoother is called the support of the smoothed value. For example $\{x_i, \dots, x_{i+n}\}$ is the support of $(\vee^n x)_i$ and the corresponding index interval, $[i, i+n]$, is called the support index interval. Hence, the window size of a smoother is defined as the number of elements in the support. The support index intervals and window sizes implied by the different smoothers, discussed in this paper, are:

- $\vee^n : [i, i+n]$, window size = $n+1$,
- $\wedge^n : [i-n, i]$, window size = $n+1$,
- U_n and $L_n : [i-n, i+n]$, window size = $2n+1$,
- $L_n U_n$ and $U_n L_n : [i-2n, i+2n]$, window size = $4n+1$,
- C_n and $F_n : [i-2n(n+1), i+2n(n+1)]$, window size = $2n(n+1)+1$,
- $B_n : [i-2n(n+1), i+2n(n+1)]$, window size = $2n(n+1)+1$.

2.2 Properties

The compound LULU smoother B_n , defined above, has a number of useful properties that are listed below. The derivations of these are readily available in the LULU literature.

Property 2.2.1

$$U_n L_n \leq F_n \leq B_n \leq C_n \leq L_n U_n.$$

Property 2.2.2

For each n , $B_n x = x$ if and only if x is n -monotone where an n -monotone sequence is defined as a sequence such that the subset $(x_i, x_{i+1}, \dots, x_{i+n+1})$ is monotone increasing or monotone decreasing for each i .

Property 2.2.3

For each nonnegative integer n , $B_n(B_n(x))_i = B_n(x)_i$, i.e. B_n is idempotent.

Property 2.2.4

For each nonnegative integer n , $(I - B_n)x - B_n(I - B_n)x = (I - B_n)^2 x$. Thus B_n is co-idempotent iff $B_n(I - B_n)x = 0$.

Definition 2.2.1

The total variation of a sequence x is defined as

$$T(x) = \sum_{i=-\infty}^{\infty} |x_{i+1} - x_i|.$$

Property 2.2.5

The LULU smoother B_n , satisfies an additive decomposition of the total variation, viz.

$$\begin{aligned} T(x) &= T(B_n x) + T(x - B_n x) \\ &= T(B_n x) + T(B_{n-1} x - B_n x) + \dots + T(x - B_1 x). \end{aligned}$$

See Conradie et al. (2005, 2006) for practical discussions of this decomposition. At each level, the additional amount of variation removed by B_n at that level can easily be calculated. Successive trimming (peeling off) of the sequence can be monitored by means of this decomposition of variation. Based on this, a decision can be made whether further smoothing is useful.

Definition 2.2.2

An n -blockpulse is a sequence x such that

$$x = \{\dots, 0, b_1, b_2, \dots, b_n, 0, \dots\}$$

with $b_1 = b_2 = \dots = b_n = b$ and infinitely many zeros on both sides. It is called upward if b is positive and downward if b is negative.

Usually a blockpulse appears in a time series when an n -blockpulse is added to a general monotone sequence. It is usually the result of an important event in the sequence, and a good smoother should deal with it in a consistent and known way. The smoother B_n deals with spiky impulsive noise and blockpulses in a very useful way, as given in the following property:

Property 2.2.6

The smoother B_n , removes impulsive noise in both directions and also removes upward and downward blockpulses of length $\leq n$ and preserves blockpulses of length $> n$.

Property 2.2.7

The smoother B_n , is neighbour trend preserving, i.e. for each sequence x ,

$$x_{i+1} \leq x_i \implies (B_n x)_{i+1} \leq (B_n)x_i$$

and

$$x_{i+1} \geq x_i \implies (B_n x)_{i+1} \geq (B_n)x_i.$$

Property 2.2.8

The smoother B_n , is difference reducing i.e. $|(B_n x)_{i+1} - (B_n)x_i| \leq |x_{i+1} - x_i|$

Properties 2.2.7 and 2.2.8 imply that B_n is so called fully trend preserving, i.e. it is neighbour trend preserving and difference reducing.

Property 2.2.9

The smoother B_n , recovers trends without noise, exactly.

Property 2.2.10

The smoother B_n , is location equivariant, i.e. adding a constant to all observations (in the window) changes the estimate accordingly and it is also scale equivariant, i.e. multiplication of all observations with a constant changes the output in the same way

Property 2.2.11

The smoother B_n , does not delay a level shift when there is an impulse in the opposite direction within k points after the shift.

See for example Figure 1 with $n = 3$, window size = 25 and a shift at time point 0, that means the estimate at time point 0 should preserve the shift despite an impulse at time point $k = 12$.

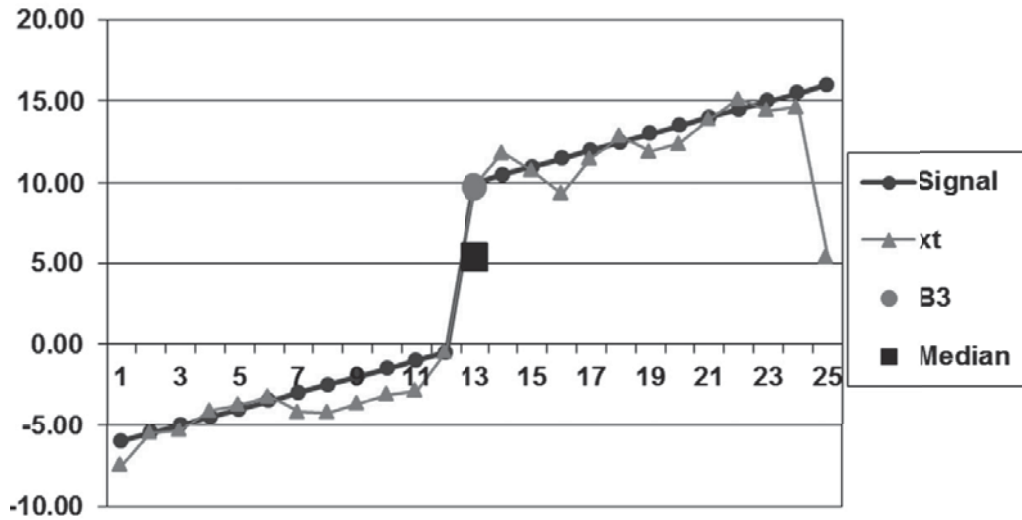


Figure 1. Property 2.2.11

The properties of B_n , discussed above, suggest that it should smooth online monitoring data under certain circumstances in a way competitive to other nonlinear smoothers. To determine how B_n compared to the running median in attenuating Gaussian noise, resisting outliers and preserving abrupt shifts in case of a constant signal and in a trend period, a simulation study was carried out and will be discussed in the following sections.

3. Simulation Study

In this section the LULU smoother B_n is compared to the running median M_k . In order to make these comparable, n and k are taken such that the respective window sizes are the same. For B_n the window size is $m = 2n(n + 1) + 1$ and for M_k it is $m = 2k + 1$. The choice of the window size m depends on the particular application one has in mind. For removing patches of subsequent outliers, the window size needs to be chosen sufficiently large.

In Fried et al. (2006) a window size of 21 was used. Choosing $m = 25$ in our case makes the results relatively comparable to Fried et al. (2006). This implies taking $n = 3$ ($m = 2 \times 3 \times 4 + 1 = 25$) and $k = 12$, thus comparing B_3 to M_{12} .

Monte Carlo experiments to compare the compound LULU smoother with the running median in basic data situations and in the presence of observational noise were carried out. The components model (1.1),

$$x_t = \mu_t + u_t + v_t,$$

was used, with

$$\mu_t = \mu_0 + \beta t$$

with the observational noise generated by the autoregressive model of order one (AR(1)), i.e.

$$u_t = \phi u_{t-1} + \varepsilon_t$$

with $\{\varepsilon_t\}$ i.i.d. $N(0, 1)$ and v_t is spiky impulsive noise from an outlier generating mechanism.

3.1 Efficiency

To compare the attenuation of Gaussian noise by B_3 and the running median M_{12} , 2 621 windows of size 25 for each slope $\beta = 0, 0.05, \dots, 0.5$, correlation $\phi = 0, 0.6$ and 0.9 and $v_t = 0$ were simulated. The efficiency of the sth

smoother, $s = 1, 2$, as measured by the percentage MSE of the arithmetic mean relative to each smoother is taken as:

$$\frac{MSE(mean)}{MSE(smoothers_s)} = \frac{\frac{1}{m} \sum_{i=1}^m (\bar{x}_i - \theta)^2}{\frac{1}{m} \sum_{i=1}^m (f_{si} - \theta)^2} \cdot 100\%, \quad f_1 = B_3 \text{ and } f_2 = M_{12}$$

with $m = 2\,621$, $\bar{x}_i = \sum_{j=1}^{25} x_{ij}$, the mean of the 25 observations in the i th window, $i = 1, \dots, 2\,621$, θ the signal and f_{si} the s th smoother value of the i th window.

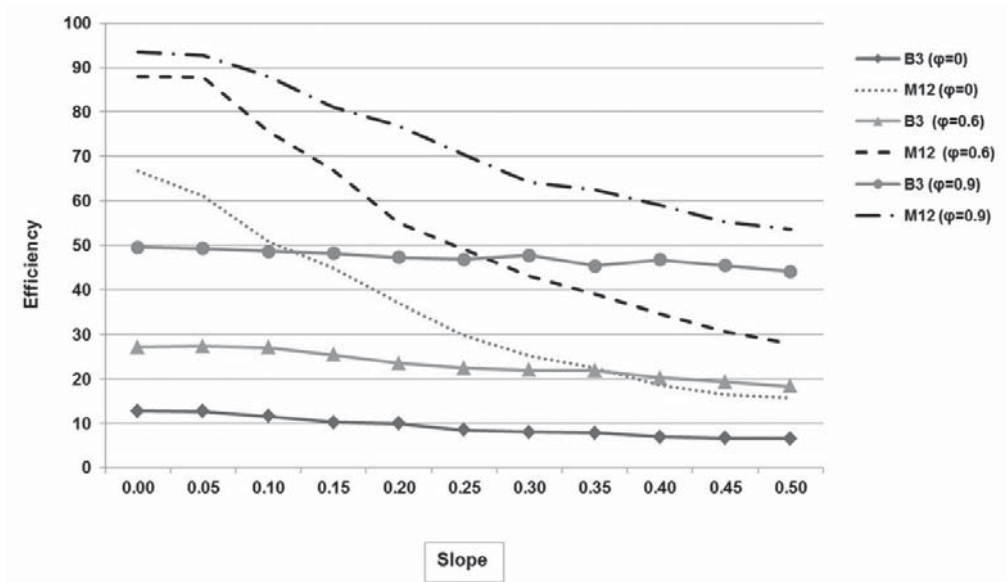


Figure 2. Relative efficiencies for Gaussian noise of smoothers for different slope and autocorrelation

The efficiencies of B_3 and the median are depicted against the values of β for $\phi = 0, 0.6$ and 0.9 in Figure 2. The standard running median M_{12} loses a lot of efficiency with increasing slope which agrees with the results in Fried et al. (2006). The relative efficiency of M_{12} increases as the autocorrelation increases for all slopes. The smoother B_3 is less efficient than the running median. However, its efficiency is less dependent on the slope; in fact the efficiency is relatively constant with increasing slope. As in the case

of the repeated median and hybrid filters of Fried et al. (2006), the relative efficiency of B_3 increases as the autocorrelation increases for all slopes with a substantially larger increase from $\phi = 0.6$ to $\phi = 0.9$ than from $\phi = 0.0$ to $\phi = 0.6$. To conclude, the efficiency of B_3 is practically independent of the slope while it is strongly dependent on the autocorrelation coefficient.

3.2 Preservation of shifts

Level shifts in a series need to be identified as soon as possible and tracked as accurately as possible. In order to investigate the preservation of a shift within a linear trend, symmetric windows of size 25 are used, estimating the level in the center of the window i.e. at time 0. Intrusion of a level shift means that it is imagined that there is a level shift starting at some point t and that the window is moved in the direction of t . When estimating the level at time $t - 12$, the window is $(t - 12) + j$, $j = -12, \dots, 12$ and ends at time t , so there is one shifted observation in the window, namely at $j = 12$, ($v_t = 10$). When estimating the level at $t - 11$, there are two outliers in the window at $j = 11, 12$ ($v_t = 10$, $v_{t-1} = 10$) etc. Ideally, the smoother output should not be affected until the window is centered at the first shifted observation. Due to symmetry this corresponds to the case where the first 13 observations in the window (including the center) are not shifted and the last 12 observations are shifted implying that there was not yet a level shift at the center of the window and the estimate of the center value should not be affected by the shift.

Figure 3 is an example of the latter.

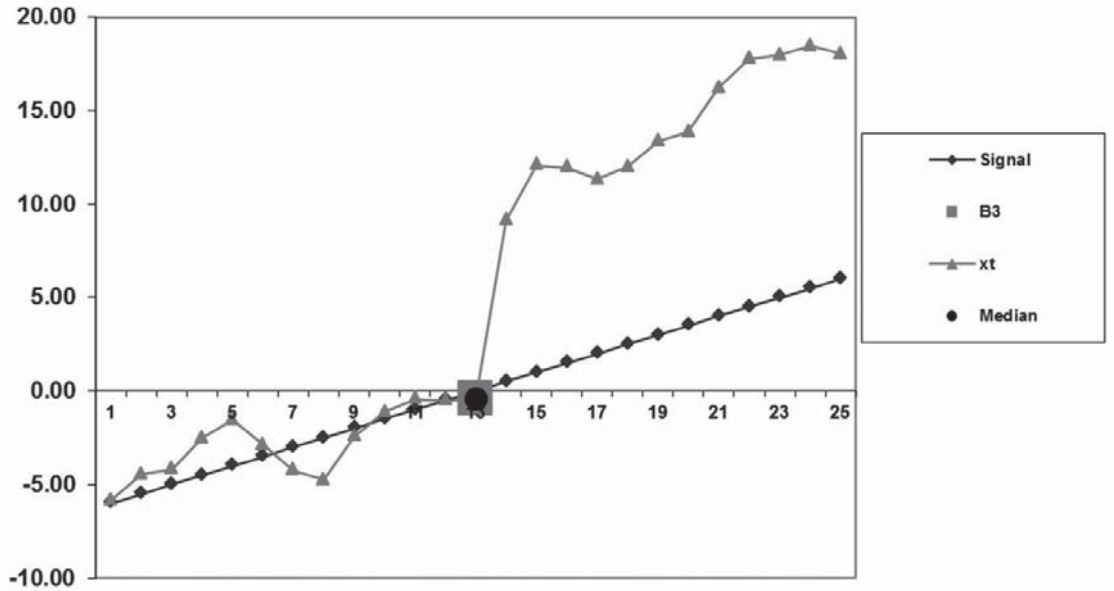


Figure 3. Example of 12 outliers to the right –
no level shift at center

Windows of size 25 were simulated using the components model, i.e. $x_t = \mu_t + u_t + v_t$, $t = -12, \dots, 0, \dots, 12$ with $\mu_t = 0 + \beta t$, $\beta = -0.5, 0, 0.5$, $u_t = \phi u_{t-1} + \varepsilon_t$, $\phi = 0, 0.6$ and $\{\varepsilon_t\}$ i.i.d. $N(0, 1)$. By replacing an increasing number (1, 2, ..., 12), of observations at the end of the window by additive outliers of size 10, level shifts were mimicked. For each setting 2 000 windows were simulated and the bias, the standard deviation and the root of the mean square error (RMSE) were calculated. In Figure 4 the RMSE is depicted against the number of outliers; shift and trend in same direction (left), shift and trend in opposite direction (right), $\beta = 0.5$ (top), $\beta = 0$ (center), $\beta = -0.5$ (bottom) and in each graph $\phi = 0$ and $\phi = 0.6$ are considered.

It was found that the RMSE of B_3 is not dominated in general by bias in contrast with the repeated median and hybrid filters of Fried et.al (2006) except in the cases of 11 and 12 outliers, when the shift and trend are in opposite directions and $\beta = -0.5$ and 0.5 . In general the magnitude of the RMSE of

the median in the different cases corresponds with calculations in Fried et al. (2006). Clear symmetric patterns are evident. The graphs for $\beta = -0.5$ and 0.5 are the same when the shift and trend position are in the same directions (left) and when the shift and trend are in opposite directions (right). Obviously when $\beta = 0$ (no trend), upward shifts (left) and downward shifts (right) imply the same situation.

The most important conclusions that follow from the simulations are:

- In the case of a trend ($\beta = -0.5$ and 0.5) and if the shift and trend are in the same direction the median performs better than B_3 for 11 or fewer outliers. In the case of 12 outliers the performance of the median and B_3 are the same. The RMSE increases for both values of β for an increase of ϕ from 0 to 0.6.
- In the case of a trend ($\beta = -0.5$ and 0.5) and if the shift and trend are in opposite directions the RMSE of B_3 stays reasonably constant at a reasonably low value for 10 or fewer outliers and then increases slightly for 11 and 12 outliers, that is when B_3 becomes more biased. The RMSE of the median increases substantially with an increase in the number of outliers. It is clear that B_3 performs substantially better than the median especially for a high number of outliers. The difference in the RMSE of the median for the two values of the autocorrelation is about 40% at 1 outlier, but then decreases fast as the number of outliers increases. For 5 or more outliers the difference is 1% or less. In the case of B_3 , the increase in the RMSE when ϕ increases from 0 to 0.6 is between 40% and 60% for $\beta = 0.5$ and between 20% and 40% for $\beta = -0.5$.

- In the case of no trend ($\beta = 0$) there is no difference whether the shift and trend are in the same or opposite directions. The RMSE of B_3 stays relatively constant for 10 or fewer outliers and then increases slightly for 11 and 12 outliers. The RMSE of the median increases with an increase in the number of outliers. For $\phi = 0$, B_3 performs better than the median for 8 or more outliers and for $\phi = 0.6$, the performance of B_3 is better for 9 or more outliers.

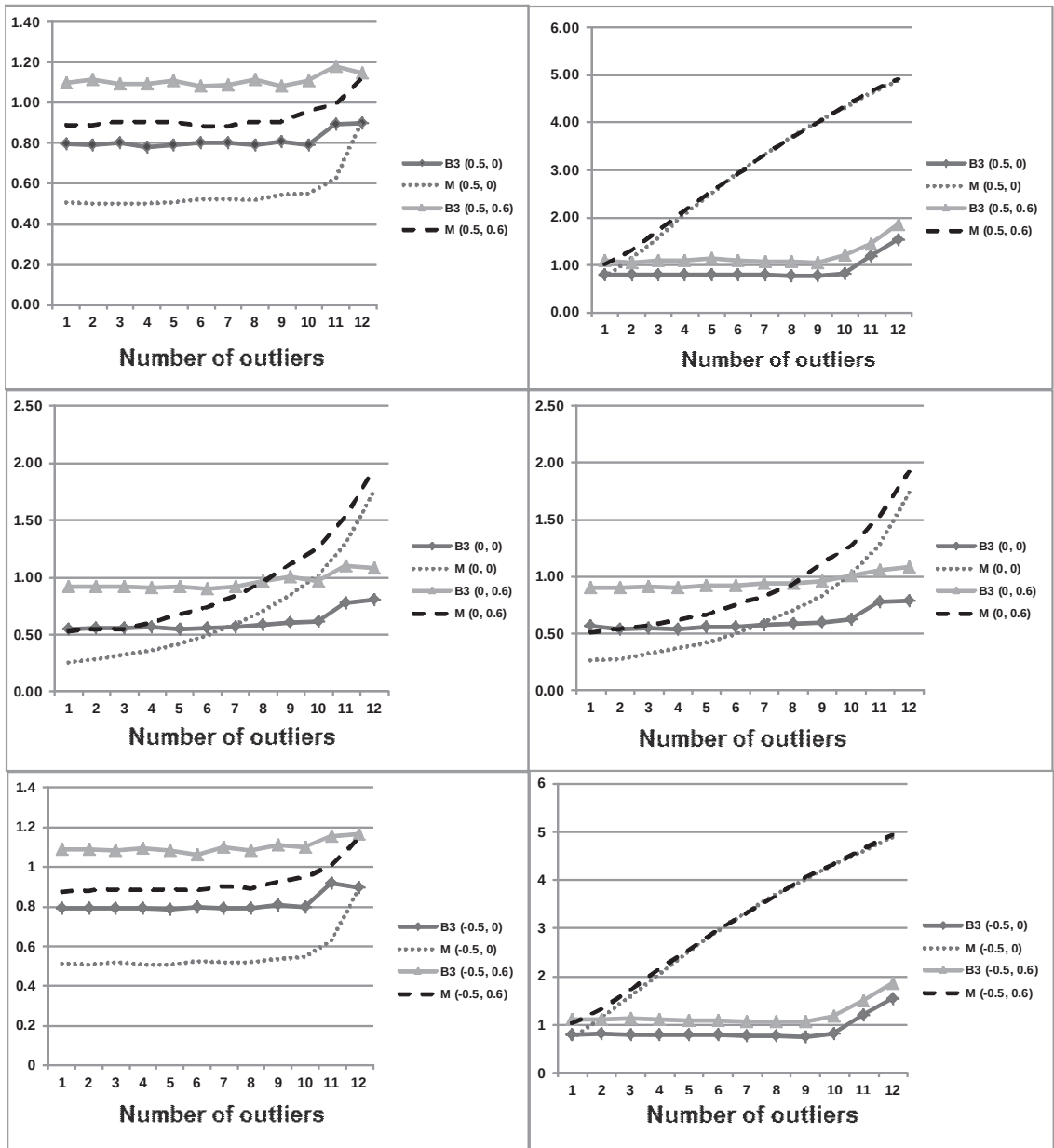


Figure 4. RMSE for the intrusion of a level shift: shift and trend in same direction (left), shift and trend in opposite direction (right), $\beta = 0$ (center), $\beta = -0.5$ (bottom) for $\rho = 0$ and 0.6

To summarise, the smoother B_3 performs substantially better than the median in the case of a trend with the shift and trend in opposite directions but, the median mostly performs better when the shift and trend are in the same direction. When there is no trend B_3 performs better when there are many outliers.

3.3 Removal of patchy impulsive noise

If patchy impulsive noise (outliers, spikes) is observed and if it does not represent a level shift it should be removed from a series. In practical applications irrelevant patches of individual outliers or several consecutive outliers of similar size, which should not influence the smoother outcome, are found. If, as in the previous section, it is imagined that a window is moved through the data, an outlier patch can be found at any location in the window at some time point. The case with one outlier patch at the start or the end of the window has been considered in Section 3.2. In this section the cases where the outlier patch starts to the right of the center and at the center, will be investigated.

In the first case an increasing number, $(1, 2, \dots, 12)$, of observations will be replaced by additive outliers (v_t) of size 10. The first situation will be a window with one outlier just to the right of the center, i.e.

$$\{x_{-12}, \dots, x_{-1}, x_0, x_1 + 10, \dots, x_{12}, \}.$$

The second situation is a window with a patch of two outliers just to the right of the center, i.e.

$$\{x_{-12}, \dots, x_{-1}, x_0, x_1 + 10, x_2 + 10, x_3, \dots, x_{12}, \}$$

etc. In all these situations the smoother outcome should not be influenced by the outliers.

In the second case an increasing number, $(1, 2, \dots, 13)$, of observations will be replaced by additive outliers (v_t) of size 10. The first situation will be a window with one outlier at the center i.e.

$$\{x_{-12}, \dots, x_{-1}, x_0 + 10, x_1, \dots, x_{12}\},$$

the second situation a window with a patch of two outliers one at the center and one just right of the center i.e.

$$\{x_{-12}, \dots, x_{-1}, x_0 + 10, x_1 + 10, x_2, \dots, x_{12}\},$$

etc. In the second case it is assumed that as soon as there are 7 or more outliers i.e. one at the center and six just right of the center, that a level shift has occurred and the smoother should pick it up. When there are 7 or more outliers it is assumed that the signal is $\mu_0 + \beta t + 10, t \geq 0$.

For the same settings of β and ϕ as in Section 3.2, 2 000 windows were simulated for the two cases discussed above and the bias, the standard deviation and the root mean square error (RMSE) were calculated.

In Figure 5 the RMSE for the first case where the outliers start just to the right of the center is depicted against the number of outliers; outliers and trend in same direction (left), outliers and trend in opposite direction (right), $\beta = 0.5$ (top), $\beta = 0$ (center), $\beta = -0.5$ (bottom) and in each graph $\phi = 0$ and $\phi = 0.6$ are considered. It was found that the RMSE's are mainly due to bias when there is no trend and the number of outliers is 3 or more and also when there is trend and the outliers and the trend are in opposite directions and the number of outliers is more than 2. When there is trend and the outliers and the trend are in the same direction the bias is very small relative to the variance and does not dominate the RMSE. Similar symmetric patterns as in Section 3.2 are evident in Figure 5.

The most important conclusions that follow from the simulations are:

- In the case of a trend ($\beta = -0.5$ and 0.5) and if the outliers and trend are in the same direction, the median performs better than B_3 for 5 or fewer outliers when $\phi = 0$ and 7 or fewer outliers when $\phi = 0.6$. For more outliers the performance of the median and B_3 are more or less the same. The RMSE increases for both values of β for an increase of ϕ from 0 to 0.6
- In the case of a trend ($\beta = -0.5$ and 0.5) and if the shift and trend are in opposite directions the RMSE's of B_3 and the median are very similar for 4 or fewer outliers and both values of ϕ . For 5 or more outliers the RMSE for the median increases substantially as the number of outliers increases while the RMSE's of B_3 stay constant with little differences in the values for the two values of ϕ and B_3 performs thus better than the median.
- In the case of no trend ($\beta = 0$) there is no difference whether the shift and trend are in the same or opposite directions. The RMSE of B_3 stays relatively constant for 3 or more outliers. The RMSE of the median increases with an increase in the number of outliers. For $\phi = 0$ and 0.6 , B_3 performs better than the median for 10 or more outliers.

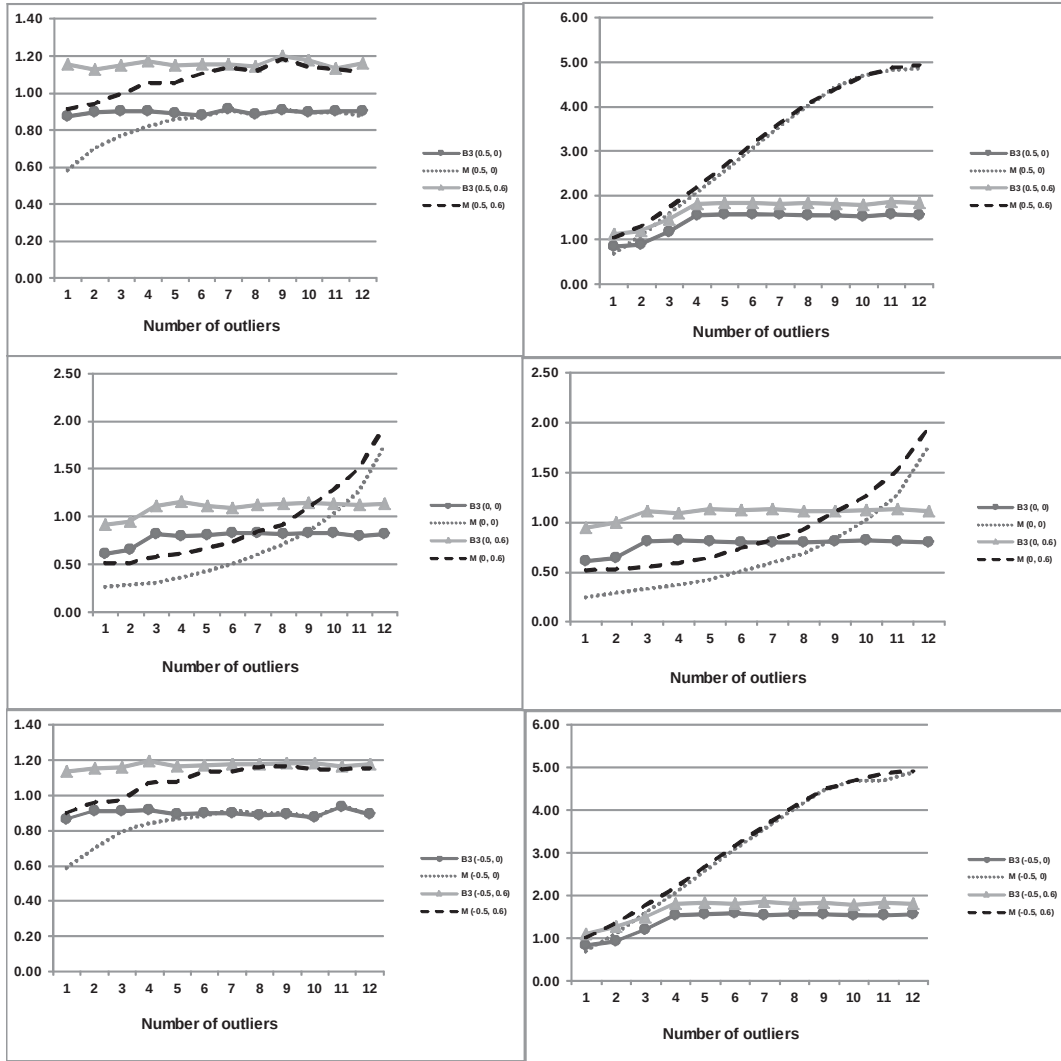


Figure 5. RMSE for patches of outliers: outliers and trend in same direction (left), outliers and trend in opposite direction (right), $\beta = 0.5$ (top), $\beta = 0$ (center), $\beta = -0.5$ (bottom) for $\rho = 0$ and 0.6

To summarise, the smoother B_3 performs substantially better than the median in the case of a trend with the outliers and trend in opposite directions and when there is no trend, B_3 performs better when there are many outliers and the median performs better when there are fewer outliers. The median and B_3 perform very similar in the case of a trend with the outliers and trend in the same direction and the number of outliers is 7 or more and for fewer outliers the median is better.

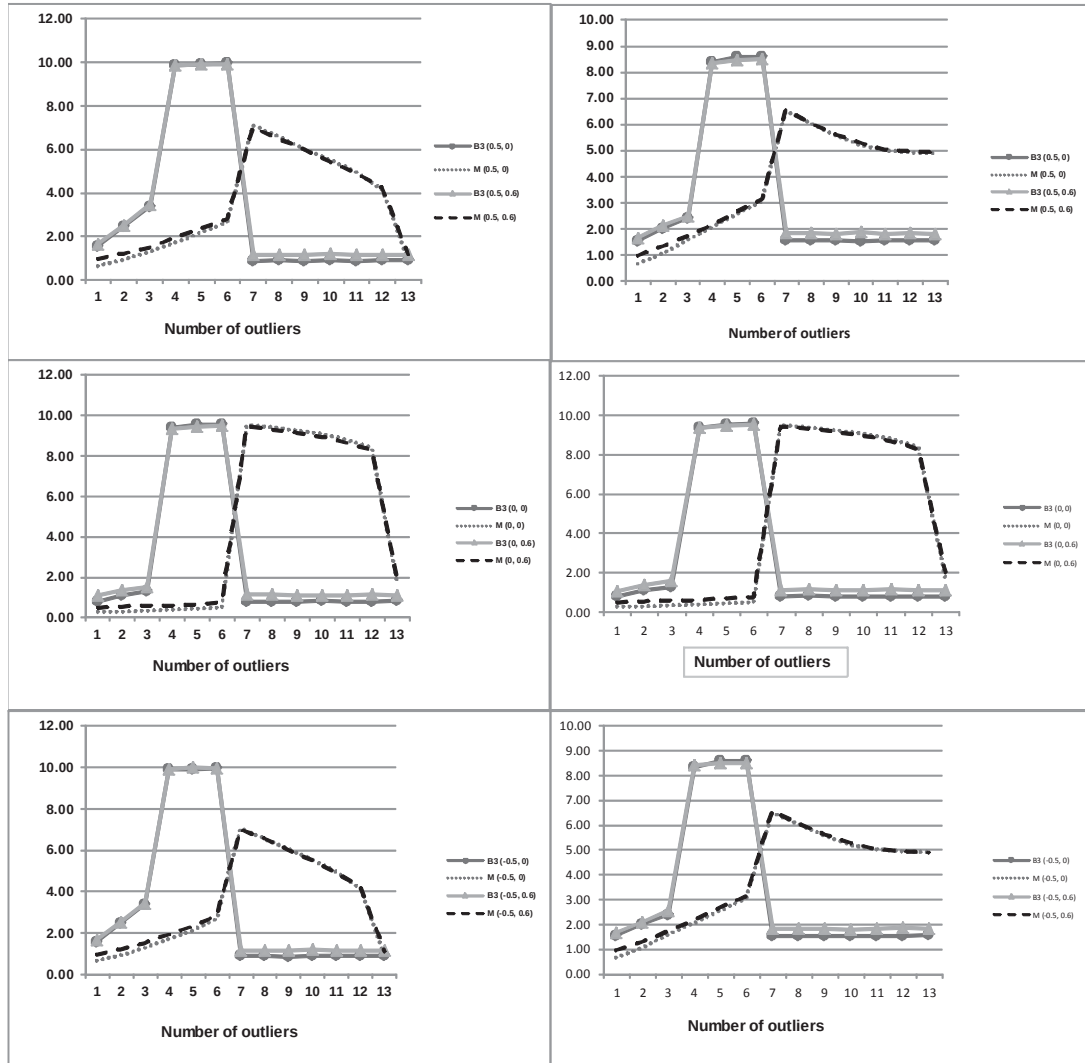


Figure 6. RMSE for patches of outliers: outliers and trend in same direction (left), outliers and trend in opposite direction (right), $\beta = 0.5$ (top), $\beta = 0$ (center), $\beta = -0.5$ (bottom) for $\rho = 0$ and 0.6

In Figure 6 the RMSE for the second case where the outliers start at the center is depicted against the number of outliers. The graphs of the RMSE of B_3 are the same for all the settings and correspond to the important property (Property 2.2.6) of LULU smoothers, that is, B_n removes impulsive noise in both directions and also removes upward and downward blockpulses of length $\leq n$ and preserve blockpulses of length $> n$. Here $n = 3$ which means blockpulses of length ≤ 3 will be removed and blockpulses of length > 3 will be preserved. That is why the RMSE is relatively low for 3 or fewer outliers since it is removed and the smoother outcome corresponds to the signal. It is high for blocks of 4, 5 or 6 outliers since these blocks are not removed by B_3 , while according to the assumptions made, a level shift has not occurred yet and hence the smoother outcome and the signal are totally different. The RMSE is low for blocks of 7 and more outliers, since these blocks are not removed by B_3 and it is assumed that a level shift has occurred if there are 7 or more outliers and hence the smoother outcome corresponds with the signal. In Figure 7 examples of simulations for blocks of 3, 4, 5, 6, 7 and 8 outliers are depicted, illustrating this argument graphically.

From Figure 6 it is clear that the median performs better for blocks of 6 or fewer outliers in this case. Due to the way the median smooths, it removes the outlier patches of lengths 7 to 12 when there is trend and the outliers and the trend are in the same direction and also in the no trend situation. When there is trend and the outliers and the trend are in opposite directions, outlier patches of lengths 7 to 13 are not removed by the median. These situations lead to high RMSE's.

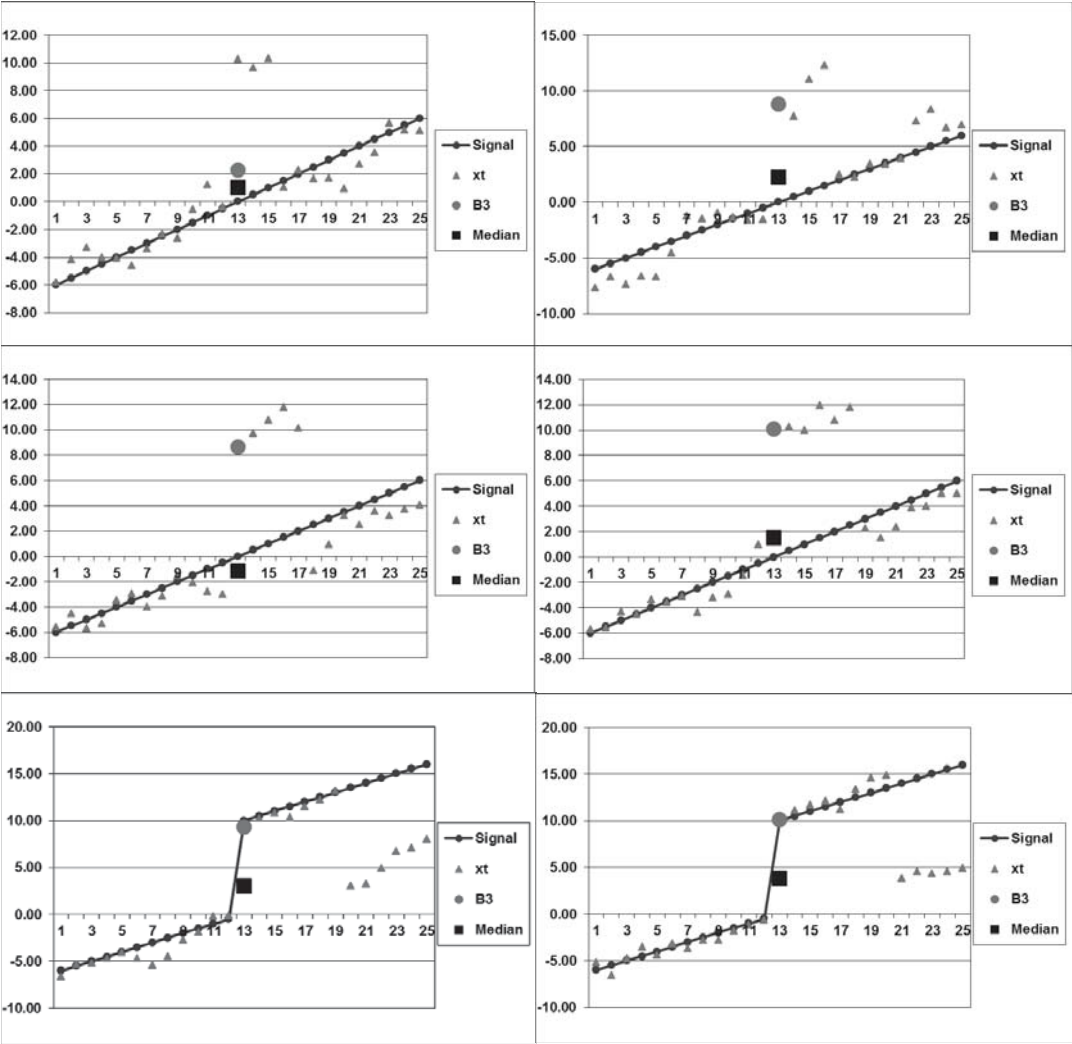


Figure 7. Examples of simulations for blocks of 3, 4, 5, 6, 7 and 8 outliers. For 7 or more outliers a level shift is assumed.

Due to the way B_3 deals with blocks of outliers and depending on how a level shift is defined regarding the number of outliers in a block and the position of the block in a window, B_3 will perform well or poorly. The fact is, it is known how B_n deals with blocks of outliers of length $\leq n$ and $> n$. From Figure 6 it is clear that if blocks of outliers of length 4 to 12, starting at the center, are present and these outliers are considered not to change the smoother output, B_3 will perform poorly. In contrast with this, if blocks of outliers of length 13, starting at the center, are present and if this indicates a level shift i.e. the smoother should pick it up, B_3 will perform very well in picking it up.

To highlight the way B_3 deals with blocks of outliers at different positions in the window, a number of examples are given in Figure 8. This can be summarised as follows:

- (a), (b) and (f): Blocks of outliers at the beginning and end of the window:

B_3 does not remove the outliers if there are four or more observations left at the center that are not outliers.

- (c) and (d): Blocks of outliers at the beginning, center and end of window:

B_3 only removes the block if it has 3 or fewer outliers.

- (e): Blocks of outliers of length 3 separated by 2 or 3 spaces: B_3 does not remove the outliers although it is blocks of outliers of size 3.

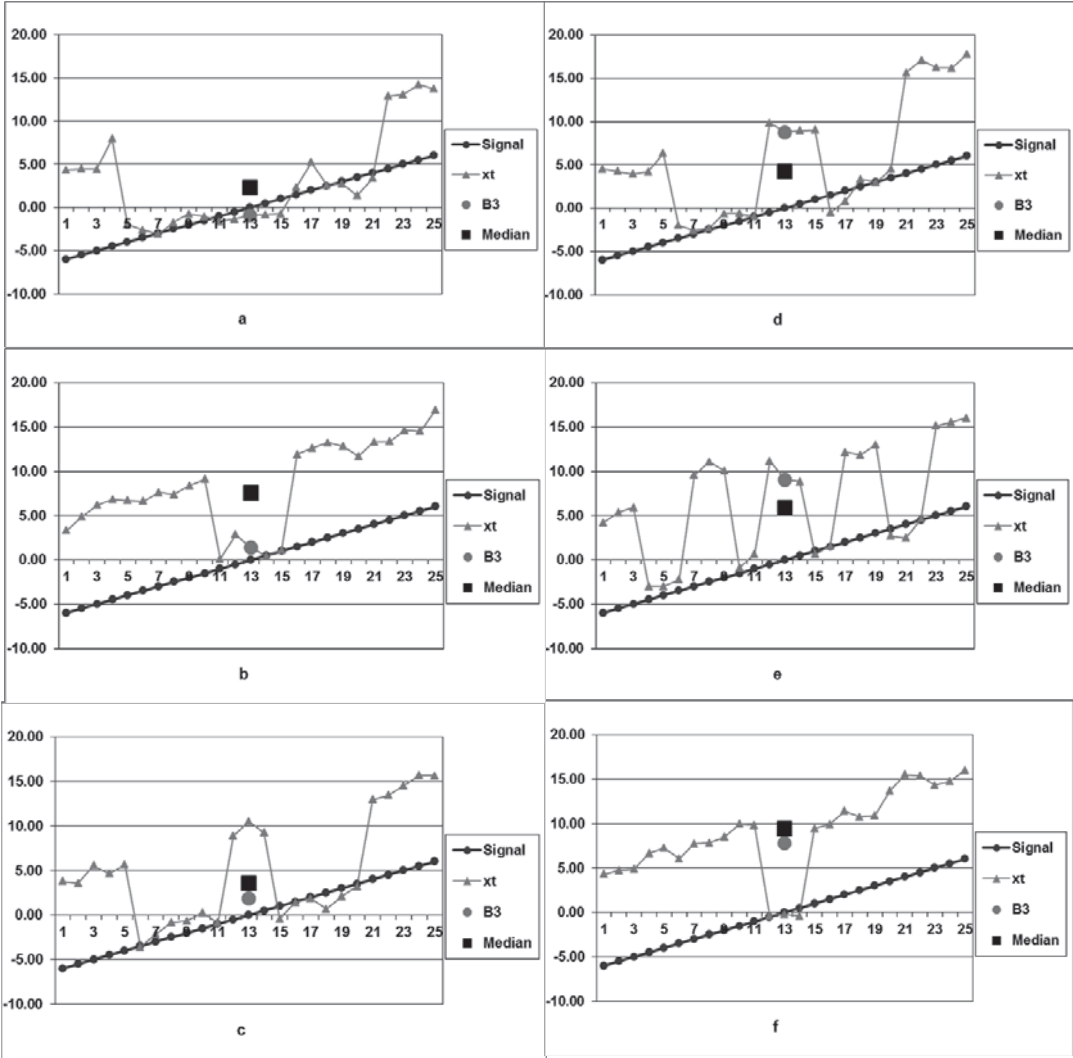


Figure 8. Examples of simulations for blocks of outliers of different lengths at different positions.

The fact that it is known how B_n deals with blocks of outliers, makes it easy to apply in a practical situation. The size of n will be determined by the definition of the size of the block of outliers that need to be removed or preserved as guided by the practical circumstances.

4. Practical examples

Fried et al. (2006) analysed time series representing arterial blood pressure using the repeated median and hybrid smoothers to extract the signal. The first time series has a blockpulse and a step edge as special features, while the second time series has a ramp edge and a few very large outliers. To investigate how the LULU smoother B_n treats these features, two time series representing two artificial arterial blood pressure series over 300 time units, with similar patterns were set up (cf. Jankowitz 2007). In the first data set the observations vary between 80 and 120 with a few outliers, then it increases to about 200 units with a number of significant outliers for about 20 time units. Thereafter it decreases to 80 units whereafter it varies following a downward slope towards a turning point at 20 units. It then increases to between 100 and 120 units with a few small and one large outlier. In Figure 9 the series, B_3 and M_{12} are pictured for example. The step-down end-value rule was applied to this example to obtain smoothed values at the ends of the series.

From Figure 9 it is clear that B_3 and M_{12} give very similar results. Both of these smoothers preserve the blockpulse and are not affected by a single impulse or outlier. The advantage that B_3 has over the median and other nonlinear smoothers is that the amount of variation removed at each smoothing level can be calculated (cf. Property 2.2.5). The decomposition of the total variation by B_n , $n = 1, \dots, 5$, as well as the corresponding percentages at each

smoothing level, are tabulated in Table 1.

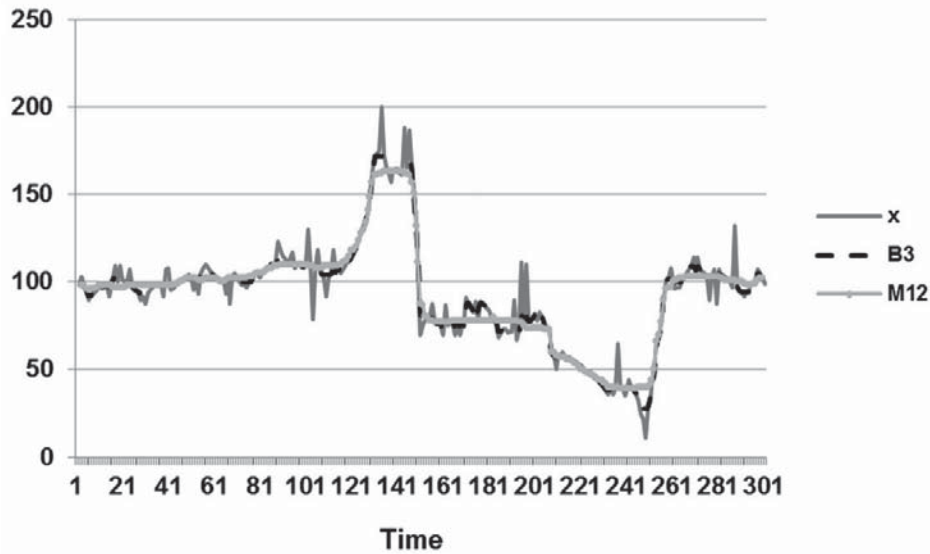


Figure 9. Series 1

The total variation of this time series is $T(x) = 1\,432$. From Table 1 follows that B_1 removes 38.69% and B_3 , 22.49%. From the five smoothers these are the two highest percentages. For the other three values of n , less than 5% of the total variation is removed when applied. Stopping after B_1 or B_3 has been applied would be good stopping choices. B_1 would be a good choice if results are needed immediately with no concern for the degree of smoothing, while B_3 will produce a smoother series, removing blockpluses of length three or less. Although the median does not satisfy the additive decomposition of the total variation, it is interesting to note that after M_2 , which has the same window size as B_1 , was applied 31.07% of the total variation was not removed in comparison with 61.38% in the case of B_1 . For M_6 and B_2 the respective percentages are 22.49% and 57.12%.

Table 1: Reduction of total variation for B_n on time series 1									
n	$T(x)$	$T(B_n)$	$T(B_{n-1}-B_n)$	$T(B_{n-2}-B_{n-1})$	$T(B_{n-3}-B_{n-2})$	$T(B_{n-4}-B_{n-3})$	$T(B_{n-5}-B_{n-4})$		
1	1.432	878	554						
%	100%	61.31%	38.69%						
2	1.432	818	60	554					
%	100%	57.12%	4.19%	39.69%					
3	1.432	496	322	60	554				
%	100%	34.64	22.49%	4.19%	39.69%				
4	1.432	463	33	322	60	554			
%	100%	32.33%	2.30%	22.49%	4.19%	39.69%			
5	1.432	446	17	33	322	60	554		
%	100%	31.15%	1.19%	2.30%	22.49%	4.19%	39.69%		

In the second data set, the observations start at approximately 62 with little variation and then increase gradually to about 94 with a few very large outlier impulses. Thereafter the observations gradually drop to 84, followed by the largest outlier of 108. It increases to 88 and stays there for a while, whereafter it drops to 78, increases again to stabilise around 90, with some large outliers when it increases. In Figure 10 the series, B_2 and M_6 are pictured.

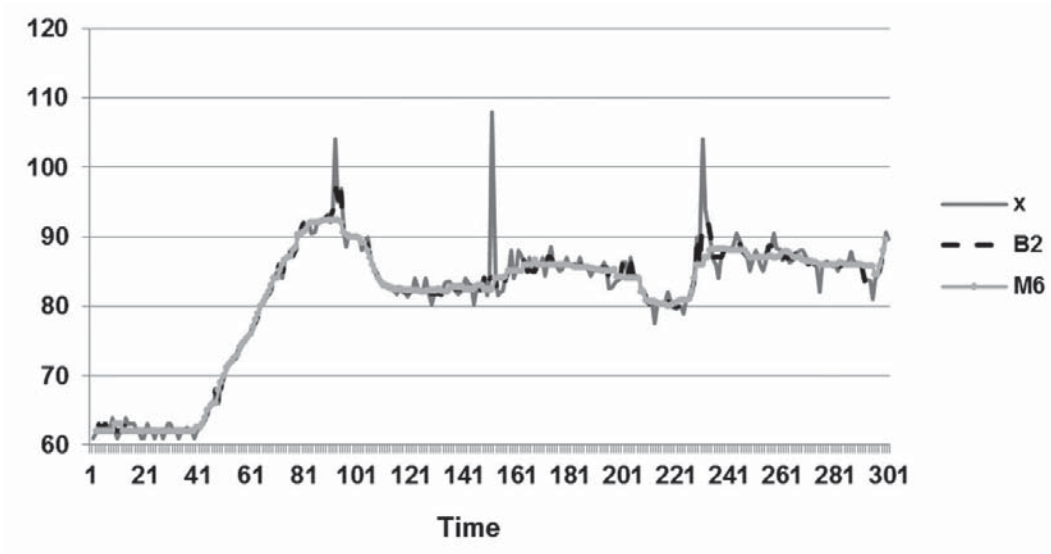


Figure 10. Series 2

The decomposition of the total variation of series 2 by $B_n, n = 1, \dots, 5$, as well as the corresponding percentages at each smoothing level, are tabulated in Table 2. The total variation of series 2 pictured in Figure 10, is $T(x) = 340.6$. From Table 2 it follows that the most variation is removed by the compound smoother B_1 (51.73%). For the other four values of n , less than 6.20% of the total variation is removed when applied. For this time series smoothing could have been stopped after applying B_1 .

Table 2: Reduction of total variation for B_n on time series 2									
n	$T(x)$	$T(B_n)$	$T(B_{n-1})$	$T(B_{n-2})$	$T(B_{n-3})$	$T(B_{n-4})$	$T(B_{n-5})$		
			B_n	B_{n-1}	B_{n-2}	B_{n-3}	B_{n-4}		
1	340.6	164.4	176.2						
%	100%	48.27%	51.73%						
2	340.6	144.6	19.8	176.2					
%	100%	42.45%	5.81%	51.73%					
3	340.6	123.6	21	19.8	176.2				
%	100%	36.29	6.17%	5.81%	51.73%				
4	340.6	109.8	13.8	21	19.8	176.2			
%	100%	32.24%	4.05%	6.17%	5.81%	51.73%			
5	340.6	105.2	4.6	13.8	21	19.8	176.2		
%	100%	30.89%	1.35%	4.05%	6.17%	5.81%	51.73%		

5. Conclusions

As was seen in other studies (Conradie et al. 2009, Fried et al. 2006) regarding the effectiveness of nonlinear smoothers, none of the smoothers is overall optimal. In this paper the following findings were made in comparing the compound LULU smoother B_n with the running median M_k with window size $2n(n+1)+1=2k+1$. In this study n is taken as 3, $k=12$ giving a window size of $m=25$.

- The smoother B_3 is not as efficient as the running median M_{12} . However, its efficiency is less dependent on the slope parameter. In fact, the efficiency is relatively constant with increasing slope. The relative efficiency of B_3 increases as the autocorrelation increases for all slopes with a substantially larger increase from $\phi=0.6$ to $\phi=0.9$ than from $\phi=0.0$ to $\phi=0.6$.
- Regarding the preservation of shifts, the smoother B_3 performs substantially better than the median in the case of a trend with the shift and trend in opposite directions, but the median mostly performs better when the shift and trend are in the same direction. When there is no trend B_3 performs better when there are many outliers.
- To study the removal of patchy impulsive noise, two cases were considered.
 - In the first case, an increasing number, $(1, 2, \dots, 12)$, of observations were replaced by additive outliers of size 10 with the first outlier just to the right of the center and the rest to the right of that one. The smoother B_3 performs substantially better than the median in the case of a trend with the outliers and trend in opposite directions. When there is no trend B_3 performs better when there are many outliers and the median and B_3 perform very similar in the case of a trend with the outliers and trend in the same direction and the number of outliers is 7 or more.

- In the second case an increasing number, $(1, 2, \dots, 13)$, of observations were replaced by additive outliers of size 10, with the first outlier at the center and the rest to the right of it and it was assumed that as soon as there were 7 or more outliers, a level shift has occurred. In this case the performance of B_3 confirms the important property of how LULU smoothers deal with impulsive noise, namely that B_3 removes impulse noise in both directions and also removes upward and downward blockpulses of length ≤ 3 and preserves blockpulses of length > 3 . Due to the way the median smooths, it performs better for blocks of 6 or fewer outliers while B_3 performs substantially better when there are 7 or more outliers.

It is clear from this study that the compound LULU smoother is a worthy competitor to the well-known nonlinear smoothers found in the literature in extracting the underlying, clinically relevant signal from an observed time series when Gaussian and spiky impulsive noise are present and also when level shifts need to be preserved. The sound theory that underlies LULU smoothers as well as their exceptional mathematical properties add to their attractiveness. In this regard the way it deals with impulse noise and blockpulses as well as the additive decomposition property of the total variation, stand out. All the indications are that LULU smoothers, in particular the compound smoother, would also compete very well with the repeated median and hybrid filters of Fried et al. (2006). Research into this is currently in progress.

References

- Conradie, W. J., De Wet, T., and Jankowitz, M. D. (2005). An Overview of LULU smoothers with Application to Financial Data. *Journal for Studies in Economics and Econometrics* 29 (1): 97-121.
- Conradie, W. J., De Wet, T., and Jankowitz, M. D. (2006). Exact and Asymptotic distributions of LULU Smoothers. *Journal of Computational and Applied Mathematics* 186: 253-267
- Conradie, W.J., De Wet T., and M. D. Jankowitz (2009). Performance of non-linear smoothers in Signal Recovery. *Applied Stochastic Models in Business and Industry* 25: 425-444
- Fried, R., Bernhold, T., and Gather, U. (2006). Repeated median and hybrid filters. *Computational Statistics & Data Analysis* 50: 2313 - 2338.
- Heinonen, P., and Neuvo, Y. (1988). FIR-median hybrid filters. *IEEE Trans. Acoust. Speech Signal Process* 35: 832 - 838
- Jankowitz, M.D. (2007). Some statistical aspects of LULU smoothers. *Unpublished PhD dissertation, Stellenbosch University, South Africa.*
- Rohwer, C. H. (1989). Idempotent One-Sided Approximation of Median Smoothers. *Journal of Approximation Theory* 58 (2): 151-163.
- Rohwer, C. H., and Toerien, L. M. (1991). Locally monotone robust approximation of sequences. *Journal of Computational and Applied Mathematics* 36: 399-408.
- Rohwer, C .H. (1999). Projections and Separators. *Quaestiones Mathematicae* 22: 219-230.
- Rohwer, C. H. (2002a). Natural Alternatives for One Dimensional Median Filtering. *Quaestiones Mathematicae* 25: 135-162.

Rohwer, C. H. (2002b). Variation Reduction and LULU-Smoothing.

Quaestiones Mathematicae 25: 163-176.

Rohwer, C. H. (2005). *Nonlinear Multiresolution Analysis*. Birkhäuser Publishers.

